Kanha Batra

Email: kbatra@stanford.edu | | LinkedIn: kanhabatra | GitHub: kanhabatra | Scholar: Google Scholar | Web: kanhabatra.github.io

Research Interests

Computational neuroscience and machine learning, with a focus on continual and meta-learning, probabilistic state-space models, neural population dynamics, and biologically-grounded architectures. Interested in bidirectional translation between neuroscience and AI: using neural circuit principles to inspire robust learning systems, and using ML methods to decode large-scale neural data.

Education

University of California, San Diego — Electrical & Computer Engineering
Ph.D. in Machine Learning & Data Science — *June 2020 – June 2024*

- Advisors: Kay M. Tye (Salk Institute) & Terrence J. Sejnowski (Salk Institute / UCSD)
- Dissertation: Probabilistic state-space models of prefrontal dynamics; neuromodulation-inspired mixture-of-experts for continual learning

University of California, San Diego — Electrical & Computer Engineering
M.S. in Machine Learning & Data Science — *September 2018 – June 2020*

Indian Institute of Technology, Delhi — Chemical Engineering
B.Tech. in Chemical Engineering — *July 2014 – May 2018*

Positions

Giocomo Lab & Linderman Lab — Stanford University — Stanford, CA
Postdoctoral Fellow — *September 2024 – Present*

- Developing biologically-grounded neural network architectures for continual learning, using hippocampal place-field remapping as a computational framework for non-stationary representation learning in AI systems.
- Building scalable state-space models (SSMs) for large-scale neural population recordings in collaboration with the Linderman lab; targeting efficient variational inference for latent dynamics in high-dimensional data.
- Designing experiments in collaboration with the Giocomo lab to probe the representational geometry of hippocampal remapping under environmental and task-context manipulations.
- Advisors: Lisa Giocomo, PhD & Scott Linderman, PhD

Systems Neuroscience Lab — Salk Institute / UC San Diego — La Jolla, CA
Graduate Student Researcher — *July 2019 – August 2024*

- Built an unsupervised probabilistic state-space model (HMM coupled with Generalized Linear Models) to decode latent behavioral states from high-dimensional mPFC population recordings during social competition; results constituted the key computational contribution to a **co-first-author *Nature* (2022)** paper (Padilla-Coreano*, Batra* et al.).
- Designed a neuromodulation-inspired **mixture-of-experts architecture** as a meta-learning framework for continual multi-task learning across non-stationary environments; validated via deep RL benchmarks and in-vivo neuroscience experiments; presented at *Cosyne 2024* and *SfN 2023*.
- Devised a novel Multi-Context "Seasonal" Foraging Task combining rodent behavioral experiments with reinforcement learning simulations to study contextual decision-making and PFC-mediated behavioral flexibility.
- Contributed probabilistic decoding analysis to the ***Nature* (2022)** study on neurotensin and valence assignment in the amygdala (Li*, Namburi*, Olson* et al., & Tye).
- Advisors: Kay M. Tye, PhD & Terrence J. Sejnowski, PhD

Seller Partner Recruitment & Success — Amazon Science — Seattle, WA
Applied Science Intern — *June 2022 – September 2022*

- Implemented Shapley Additive Explanations (SHAP) to quantify feature importance and identify high-impact events in temporal classification tasks on RNNs, directly improving model interpretability for seller recruitment decisions at production scale.
- Deployed TimeSHAP for event attribution in recurrent models; pipeline informed downstream feature engineering strategy for the seller scoring model.

Multimodal Imaging Lab — ACTRI, UC San Diego — La Jolla, CA
Graduate Student Researcher — *December 2018 – June 2020*

- Developed an ML classifier (restriction spectrum imaging combined with supervised learning) to characterize and automatically segment bone metastases in prostate cancer patients; contributed to two peer-reviewed publications (*Scientific Reports* 2022; *J. Magnetic Resonance Imaging* 2021).
- Advisors: Tyler Seibert, PhD & Anders Dale, PhD

Centre for Advanced Research in Imaging, Neuroscience & Genomics — New Delhi, India
Scientific Research Intern — *May 2018 – July 2018*

- Built a pipeline to generate 3D mesh models of the coronary system from chest CT scans and computed Fractional Flow Reserve to non-invasively quantify stenosis severity in Coronary Artery Disease patients.

Medical Image Processing Lab — IIT Delhi New Delhi, India
Undergraduate Student Researcher *January 2018 – May 2018*

- Processed EMG signals from a robotic exoskeleton for neuro-muscular rehabilitation; applied dimensionality reduction to extract features correlated with clinical spasticity scores.
- Advisor: Amit Mehndiratta, PhD

Microfluidics & Nanoscience Lab — deMello Group, ETH Zürich Zürich, Switzerland
Undergraduate Student Researcher *June 2017 – February 2018*

- Developed a C++ controller (Arduino/Atmega2560) to automate valve actuation in a microfluidic chip for cell-division experiments; built a cascade classifier for real-time detection and tracking of dividing cells in microscopic video.
- Advisor: Casadevall i Solvas, PhD

Skills

- **Frameworks & Libraries**: PyTorch, TensorFlow, JAX, Hugging Face Transformers, scikit-learn, NumPy, SciPy, pandas, wandb, SSM (state-space modeling)
- **Methods**: Continual/lifelong learning, meta-learning, deep reinforcement learning, transformer architectures, probabilistic graphical models (HMM, GLM), variational inference, Bayesian optimization, mixture of experts, SHAP/XAI, multi-task learning, dimensionality reduction
- **Programming & Infrastructure**: Python, C++, SQL, MATLAB, AWS, Spark, Git, Docker, SLURM/HPC clusters, Linux
- **Neuroscience & Experimental**: Large-scale calcium imaging analysis, fiber photometry, behavioral quantification (SLEAP/DeepLabCut), spike sorting, in-vivo electrophysiology pipelines

Publications

- Austin A. Coley, **Kanha Batra**, Jeremy M. Delahanty, Laurel R. Keyes, Rachelle Pamintuan, et al. & Kay M. Tye. Predicting Future Development of Stress-Induced Anhedonia From Cortical Dynamics and Facial Expression. **Preprint, 2024**.
- Reesha R. Patel, Makenzie Patarino, Kelly N. Kim, . . . , **Kanha Batra**, . . . , Kay M. Tye. Social isolation recruits amygdala-cortical circuitry to escalate alcohol drinking. **Preprint, 2024**.
- Chris A. Leppla, Laurel R. Keyes, Gordon Glober, Gillian A. Matthews, **Kanha Batra**, et al. & Kay M. Tye. Thalamus sends information about arousal but not valence to the amygdala. ***Psychopharmacology*, 2023**.
- Hao Li*, Praneeth Namburi*, Jacob M. Olson*, . . . , **Kanha Batra**, . . . , Kay M. Tye. Neurotensin orchestrates valence assignment in the amygdala. ***Nature*, 2022**.
- Nancy Padilla-Coreano*, **Kanha Batra***, Makenzie Patarino, Zexin Chen, Rachel R. Rock, Ruihan Zhang, et al. & Kay M. Tye. Cortical ensembles orchestrate social competition through hypothalamic outputs. ***Nature*, 2022**. *(*co-first authors)*
- Leonardino A. Digma, Christine H. Feng, . . . , **Kanha Batra**, . . . , Tyler M. Seibert. Correcting B0 inhomogeneity-induced distortions in whole-body diffusion MRI of bone metastases. ***Scientific Reports*, 2022**.
- Christine H. Feng, Christopher C. Conlin, **Kanha Batra**, Ana Rodriguez-Soto, et al. & Tyler M. Seibert. Voxel-level Classification of Prostate Cancer on MRI: Improving Accuracy Using Four-Compartment Restriction Spectrum Imaging. ***Journal of Magnetic Resonance Imaging*, 2021**.

**Authors have contributed equally*

Select Abstracts in Scientific Conferences

- **Kanha Batra** et al. & Kay M. Tye. Neuromodulated mixture of experts: A prefrontal cortex inspired architecture for lifelong learning. ***Cosyne*, Lisbon, Portugal, March 2024**.
- **Kanha Batra** et al. & Kay M. Tye. Neuromodulation as a mechanism for lifelong learning in the prefrontal cortex. ***SfN*, Washington DC, November 2023**.
- **Kanha Batra***, Nancy Padilla-Coreano* et al. & Kay M. Tye. Cortical ensembles decode social rank and promote dominance behavior via hypothalamic projections. ***SfN*, Virtual, November 2021**.
- Nancy Padilla-Coreano*, **Kanha Batra*** et al. & Kay M. Tye. A cortical-hypothalamic circuit decodes social rank and promotes dominance behavior. ***Cosyne*, Virtual, February 2021**.
- Nancy Padilla-Coreano*, **Kanha Batra*** et al. & Kay M. Tye. Prefrontal-Lateral Hypothalamic Neurons Encode Social Dominance in Mice. ***ACNP*, Virtual, December 2020** (selected for hot topic presentation).
- **Kanha Batra***, Nancy Padilla-Coreano* et al. & Kay M. Tye. Using unsupervised machine learning to understand the role of mPFC in social dominance in mice. ***Cosyne*, Denver, CO, February 2020**.

**Authors have contributed equally*

Professional & Community Service

- 2023 Organized a workshop titled "A wrinkle in time: Neuroscience at multiple timescales" at the *Computational & Systems Neuroscience Conference (Cosyne), Montreal, Canada*.
- 2021 Mentored a high school student through a summer research internship as part of Salk's Heithoff-Brody High School Scholars Program.
- 2021 Panelist representing the Systems Neuroscience Lab for Salk's Virtual March of Dimes High School Science Week.
- 2020 Participant/Volunteer, International Women's Day Wikipedia-Edit-a-thon.
- 2016–17 Chief Editor, Board for Student Publications (student-run media body, IIT Delhi).

Academic Referees

- **Lisa M. Giocomo**
  Professor, Dept. of Neurobiology, Stanford University School of Medicine
  lgiocomo@stanford.edu giocomolab.weebly.com
- **Kay M. Tye**
  Professor, Salk Institute for Biological Studies
  Adjunct Professor, Dept. of Neurobiology, UC San Diego
  tye@salk.edu tyelab.org
- **Nancy Padilla-Coreano**
  Assistant Professor, Dept. of Neuroscience, University of Florida
  npadillacoreano@ufl.edu padillacoreanolab.com
- **Terrence J. Sejnowski**
  Professor, Salk Institute for Biological Studies
  terry@salk.edu cnl.salk.edu